



RV College of Engineering®

Mysore Road, RV Vidyanikethan Post,
Bengaluru - 560059, Karnataka, India

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RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

VII Semester B.E. Regular Examination February 2026

COMMON TO AIML / CD

Stream Processing and Analytics

Time : 3 Hours

Maximum Marks : 100

Instructions to the students

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	What is a stateless design in streaming systems? Give an example with respect to live video analytics service	02	2	2
1.2	A warehouse system sends an order update and only needs confirmation that the message was received, not a processed response. Which interaction pattern applies here, and how does it improve reliability?	02	3	3
1.3	A sensor gateway crashes, but all upstream sensor nodes remain operational and can resend data. Which message logging technique is most suitable, and why?	02	3	3
1.4	Diagrammatically represent the Apache Flink architecture	02	3	2
1.5	How does message auditing and duplicate detection improve data quality.	02	2	2
1.6	What is the purpose of In-Memory Databases (IMDB)?	02	3	2
1.7	Illustrate the concepts of consumers and consumer groups in Kafka using a suitable example.	02	2	2
1.8	The three primary methods of sending messages using Kafka Producer are: _____, _____ and _____	02	1	1
1.9	To achieve at-least-once delivery semantics in Apache Kafka, the producer should set acks = _____, configure retries _____, and keep enable.idempotence = _____.	02	1	1
1.10	List any two examples of Kafka Streams	02	3	1

Part B

Question No	Question	M	CO	BT
2a	Describe the streaming pattern for collecting data with an example.	08	2	2
2b	With the help of neat diagram discuss in details the Publish/subscribe pattern for collecting data.	08	2	2
	A company owns thousands of Coke vending machines and would like to make them social. They want the machines to tweet and send push notifications with special offers to consumers who are geographically close. As if that weren't enough, there's a twist: If the closest vending machine doesn't have stock to offer, it should recommend the next closest vending machine that can offer the customer a deal. Given the scenario, answer the following queries and discuss them in detail			
3a	(i) What would be the impact to the business if the communication between the collection tier and analysis tier were interrupted for an extended period of time? (ii) How many days' worth of data can the business tolerate losing? (iii) Would you anticipate that the business would need to store historical data? (iv) Identify and justify the type of message delivery semantics does the streaming system need	08	2	2
3b	Examine the fault-tolerance challenges in the message queuing tier of a streaming system. Discuss how data reliability is affected when (i) a broker crashes, (ii) inter-broker network connectivity is disrupted, and (iii) a storage drive failure occurs.	08	2	2
OR				
4a	A large professional networking platform needed to process millions of user activity events per second—such as profile views, clicks, job searches, and feed interactions—in near real time. The goals were: · Low-latency personalization of user feeds · High throughput and scalability · Strong fault tolerance and state consistency Implement Apache samza streaming architecture to achieve the above goals. Draw a neat diagram and explain the steps in detail	10	3	3
4b	With a neat diagram, elaborate the working of the producer, the broker, and the consumer with a message queuing layer	06	2	2

Imagine that we operate a nationwide network of smart treadmills, each equipped with sensors that continuously emit data such as current running speed, incline level, heart rate, machine temperature, and power consumption. From this stream of data, we want to analyze two key metrics.

5a First, we want to compute the total energy consumption of all treadmills every 1 minute. Each minute should produce one independent result, and the calculation should not overlap with the previous minute's computation. For now, we want a global metric, though in the future we might split the results by gym location. 10 3 1

Second, we need to detect if any treadmill exceeds a critical temperature threshold within a 1-minute interval, so we can trigger preventive maintenance alerts. Each evaluation should reset after every minute, ensuring clean, non-overlapping analysis periods

Identify and justify the type of window that can be applied to resolve the problem. Elaborate the solution in detail.

5b Analyze the role of summarization techniques in stream processing with an example, and classify the different summarization techniques. 06 1 2

OR

6a Analyse the differences between sliding window and tumbling window based on its characteristics and use cases? 06 2 4

6b List the summarization techniques and elaborate any two using an example. 10 2 2

7a Discuss Commits and Offsets in Kafka. Describe any two offset commit strategies supported by Kafka. 10 2 1

7b Explain the three primary methods of sending messages through kafka. 06 2 1

OR

Assume you have a kafka topic named patient_records that contains 6 partitions (P0-P5). A consumer group with two consumers (C1 and C2). Assuming kafka uses RangeAssignor strategy, answer the following questions:

8a (i) Explain how partitions will be assigned between C1 and C2. 06 3 4

(ii) What issue can arise if you add a third consumer (C3) to the group?

(iii) Give the solution to resolve the issues

8b A large-scale e-commerce platform processes millions of online orders and payment events every day to ensure secure transactions and detect suspicious activities in real time. The platform uses a stream processing system to analyze customer behavior, payment patterns, and delivery events. To maintain 10 3 3

reliability, the company wants to implement fault-tolerant mechanisms for its streaming pipeline.

What are the potential challenges that may arise due to failures in stream processing?

How do different stream processing tools handle communication failures?

What happens if a message broker crashes?

A large retail organization operates multiple operational systems, including a MySQL database for orders, a PostgreSQL database for customer information, and a cloud-based data warehouse for analytics. The organization wants to build a real-time data pipeline to continuously move data from these systems into Apache Kafka so that downstream teams can perform analytics, reporting, and monitoring.

9a	i) Explain why Kafka Connect is an appropriate choice for this data-ingestion requirement.	10	3	3
	ii) Describe the key features of Kafka Connect that support scalability and fault tolerance in this scenario.			
	iii) Identify two limitations of Kafka Connect and explain situations where a custom Kafka Producer–Consumer approach would be more suitable.			
9b	Imagine a system where logs are generated from multiple microservices and need to be processed and stored for analysis. Illustrate a kafka pipe line for the given scenario.	06	3	3

OR

10a	“Processing with local state refers to maintaining state (data) locally within a stream-processing task or operator to keep track of past events while processing a continuous data stream.” Elucidate the statement with a neat diagram.	08	2	3
10b	Analyze the criteria for choosing between Kafka Connect and custom Kafka Producers and Consumers while designing a Kafka-based data pipeline. Give relevant example	08	3	4